

Bluetooth

The simplest and most common way for two handheld devices to communicate with each other is via Bluetooth. Bluetooth networking transmits data via low-power radio waves. It communicates on a frequency between 2.402 GHz and 2.480 GHz. This frequency band has been set aside by international agreement for the use of industrial, scientific and medical devices (ISM). Bluetooth devices send out weak signals of about 1 milliwatt to avoid interference with other devices such as computers and cell phones. This low power, however, limits its range to 10 metres, but its signal can pass through walls so direct line of sight is not required.

Another great advantage of Bluetooth is that it can connect up to eight devices simultaneously without interference. It uses a technique called “spread-spectrum frequency hopping” which, put in simple words, means it keeps changing its frequency in the designated range and thus interference, if any, lasts for a fraction of a second and goes unnoticed.



Bluetooth logo

Like any wireless network, security is a concern with Bluetooth. The automatic nature of the connection which makes it easy to use can also be used against you by sending you data without your permission. To avoid this, Bluetooth offers several security modes, and device manufacturers determine which mode to include in a Bluetooth-enabled gadget. When any other device tries to establish a connection to the user's gadget, the user has to decide whether to allow it. If you use the device quite often, you can mark the device as “trusted” and it will be able to exchange data without permission.

In spite of all the security measures, there are certain Bluetooth specific problems.

- ▶ **Bluejacking:** Bluejacking is sending text messages or audio/video files to a person without his permission. Although harmless, it can make the user think that his phone is malfunctioning.
- ▶ **Bluebugging:** Bluebugging is a more serious problem. It allows hackers to use your phone's features, for example by placing calls or sending messages without the user realising it.
- ▶ **Car Whisperer:** Car Whisperer is a piece of software that allows hackers to send audio to and receive audio from a Bluetooth-enabled car stereo.